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FABRICATE 2020
MAKING RESILIENT ARCHITECTURE

JANE BURRY / JENNY SABIN / BOB SHEIL / MARILENA SKAVARA

The Bartlett School of Architecture
UCL
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Cornell AAP
Architecture Art Planning
SWINBURNE UNIVERSITY OF TECHNOLOGY

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LITTLE ISLAND
SCULPTURAL STRUCTURE THROUGH DIGITAL DESIGN AND FABRICATION

YONG-WOOK JO / DAVID FARNSWORTH / JACOB WIEST
ARUP
ROSARIO GALLO
ACCESS ANVIL / SCOTT SYSTEM

Project Introduction and Challenges

Project Vision
Little Island is an artificial island that is being built on the Hudson River, west of Manhattan, to provide a unique park experience for the New York public. It will be owned and maintained by the public agency, Hudson River Park Trust, and be operated by Little Island, a special organisation established for the park.

Over the years, Hudson River Park Trust has successfully renovated old piers into public parks, recreational facilities or office and retail complexes. Little Island, a completely new structure on the river, will revitalise this section of Hudson River Park with almost three acres of new public park space featuring lush greenery and a diverse array of world-class arts, educational and community programming (Little Island, 2019).

Under the overall design management by the prime consultant Mathews Nielsen Landscape Architect and in collaboration with the U.K.-based designer Heatherwick Studio, Arup was selected to realise the project vision for most engineering and consulting disciplines. Standard Architects is the executive architect.

Fort Miller Company (FMC) is the precast concrete manufacturer; Tymetal, a sister company of FMC, is the steel fabricator; and Access Anvil / Scott System, another sister company of FMC, is the foam formwork fabricator and smooth formliner manufacturer.

Design Features and Challenges
The signature design of Little Island, featuring irregular, unique and complex undevelopable curved surfaces with few repetitions, sets it apart from other pier structures, but also imposes the most significant challenges for design, documentation, fabrication and erection. The geometry was not possible or practical to convey via conventional 2D documentation, so the team adopted an open approach to 3D modelling where we quickly realised that the only way to ensure geometry of construction would align on site would be a common design and construction model with the use of direct-to-manufacturing digital fabrication techniques for all the complex geometry elements.

Design

Architecturally Exposed Structural Precast Concrete Structure
Construction on the Hudson causes challenges because

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of the marine environment which typically push designs towards prefabrication to remove the need for falsework. Pier construction generally consists of piles placed in the river bed, topped with precast pile caps and planks, and tied together with a cast-in-place concrete slab. These constraints made precast the right choice for this project, and the architect desired to use structural precast elements that would be exposed with architectural quality finishes. The precast technology would need to push past current convention to meet this vision. Also, a unique structural system has been proposed which is structurally effective, constructible and, at the same time, visually intriguing.

Pots

The deck structure consists of a unique module called 'Pots', tulip-shaped assembly of precast concrete elements. Typically, a pot is 4.5m tall and approximately 37m² on plan area. A pot is supported by a 900mm diameter cylinder pile.

Each pot is composed of four to six 'petals', with thickened side struts and bottom edge beams, that sit on a central 'column head'. The column head seamlessly transitions from the pot's bulbous curved shape to the cylindrical pile geometry. The centre of the column head acts as a vertical support for planks and 'star beams' above, which span radially from the centre to the pot corners. Petal interiors typically transition from sloped to vertical moving away from the column head. Stainless steel embedded plates and welded external plates connect the petals to each other and the column head.

Pot Geometry Generation

The outside surface geometry of the pot is generated by the architect through a parametric script. Base rules of geometry are defined through discussions between architect and engineer to balance complexity with architectural vision. For example, the petal outer surface generally has a one-way curved surface except edges where it rolls up to make the petal joint smooth, so the horizontal rebars can be straight with splicing edge bars that are hooked and anchored in the cages of the strut bars. Internal surfaces are generated by Arup using another set of parametric algorithms added to the architect's scripts to meet the required structural criteria and are geometrically simple to facilitate the conventional formwork fabrication for the interior surfaces. The joint between the column head and the petals is also defined making sure the connection using steel plates is feasible and the drainage of incidental water inside the pot is considered.

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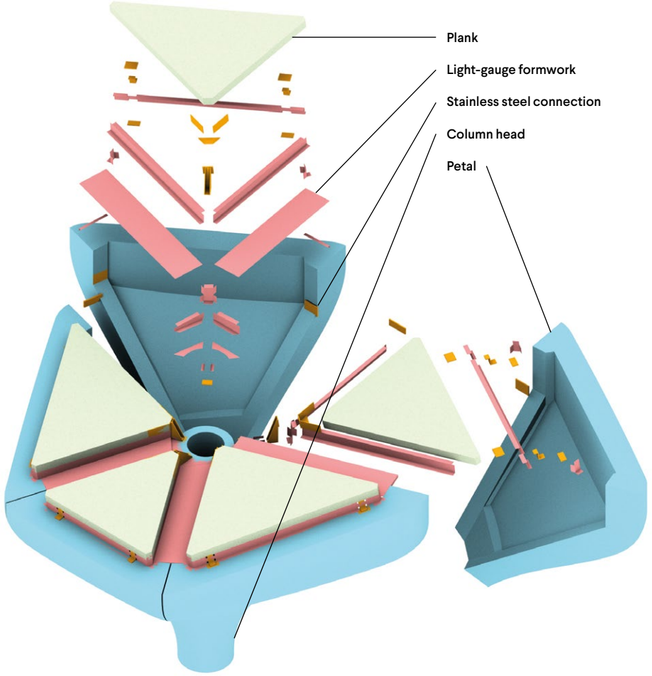

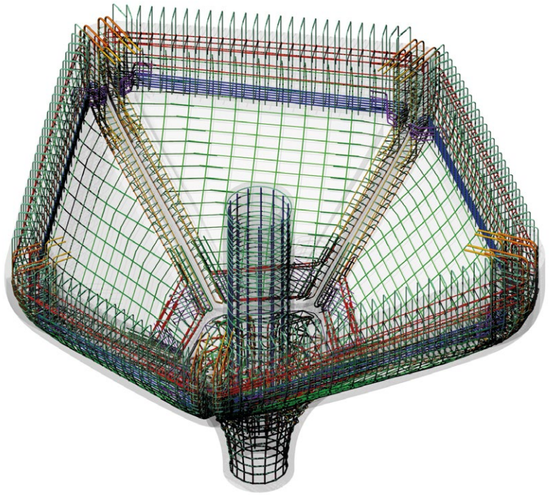

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Rationalisation

Although the overall layout and the plan shape of pots follow a specific pentagon pattern called the Cairo-Pentagon, fitting them into an overall square plan with relatively straight edges and providing varying elevations across the top surface resulted in many different pot shapes. 39 different pot types resulted in 39 column head forms and 192 unique curved petal forms used to generate 132 pots and 656 petals. The curved exterior faces at the bottom portion of the pot are typical within pot types and the elevation variation required to meet the topography of the pier are achieved through pouring varying height 'extension walls'. So, while each pot and each petal are unique, repetition of formwork was possible to a great extent. A few very special and complex pot types, particularly in areas over the southern access way and near the amphitheatre, are only used once.

Structural Analysis Model Generation

On top of the scripts that generated pot geometries, another set of algorithms is added to generate a highly-detailed analysis model in GSA, needed to design the rebar for all precast elements as well as connection plates and embeds. All design elements are explicitly modelled and custom final member design post-processing tools were created to calculate the required rebars, steel plate sizes, and the number of studs, using model force outputs directly and quickly to generate the results.

Stainless Steel Connections and Light-Gauge Sheet Metal Formwork Geometry Generation

Pot precast elements feature seven distinct connection types with multiple instances of each type based upon petal and plank geometry. This resulted in an average of 35 pieces that may have unique set-out, unique geometry, and unique plate size and weld prep definitions, dependent upon overall pot geometry and loading. Over 15,000 plates are defined and set-out parametrically. The light-gauge sheet metal formwork features 12 distinct elements to form the star beam elements. Slab geometry, which drives plank and star beam geometry, is essentially unique across all pots resulting in over 7,700 elements in total that are defined and set-out parametrically. The light-gauge includes bolt hole locations and cut-outs to avoid conflict with the steel connections.

Rebar Detailing Model Generation

Also generated in 3D using parametric scripts are the rebar models for petals and column heads. A completely detailed model includes all information needed for cutting and bending, including the clear cover, development and splice length, bending radius, etc. This was the major enabler for the realisation of the complex precast elements.

Deliverables in 3D

All surfaces of precast elements, exact shape and dimensions of stainless steel components, bent shape of light gauge stay-in-place formwork steel, and rebars in the complex petals and column heads are completely generated by the scripts with little manual adjustment. The resulting Rhino models, both the 'as-in-place' format in which all components are in their final locations in 3D, and the 'arrayed' format in which all components are laid out on a 2D surface, are provided to the precast and steel fabrication contractors with full geometry for use in shop drawing production and final fabrication.

1. Pier 55 during Construction. Photo: Timothy Schenck.

2. Exploded view of a 'Pot'.

3. Detailed 3D rebar model.

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Coordination in 3D Environment
All design team members worked in 3D using either Rhino or other BIM software, and coordinated trades in 3D as well. A Navisworks model was created by importing each team member's 3D components for coordination, clash detection and for contractor's information and reference.

Fabrication

Digital Fabrication
The fabrication team implemented new production processes to accommodate the 3D delivery of the construction documents in order to produce the complex geometric shapes. In an environment where tape measures, people, wood and steel are the norm, the utilisation of CAD/CAM technology, automatic equipment and script writing was a very novel concept that had to be implemented in order to fabricate the unique shapes within tolerance, work within a budget and meet the timeline of the project.

Fabrication Challenges
The doubly curved surfaces of the unique shapes presented the most significant challenge on the project. For the formwork, traditional mediums like wood and steel do not bend in doubly curved directions, urethane formliner lead times exceeded 48 months, carpentry requirements far exceeded current capacity and the formwork construction could not be quality controlled. Through the implementation of CAD/CAM subtractive manufacturing techniques in a production sequence, fabrication methodologies were created to build the formwork. The process included reverse engineering of the models and scripts generated during the design process for digital input of the geometry within CAD software to create the negative shape. Once created, the parts underwent a digital design process.

Model to as Built
The employment of digital fabrication for formwork production of complex architectural shapes allowed us to bridge the gap between architectural concept and the built environment. To build the formwork for curved concrete elements, CAD/CAM subtractive manufacturing techniques, utilisation of foam and polyurethane as new production mediums, and the employment of automatic equipment in a production sequence were the only means for fabrication. Large foam billets were cut on a CNC wire cutter into profiles to maximise yield of material and to fit within the parameters of the CNC five-axis milling machine. Once cut, some assembly and prep work to the foam was required prior to placement on the CNC router.

Programming Methods
The engineering process included utilising CAM software programming to direct and control the CNC cutting sequence and part verification software to ensure part fabrication feasibility prior to reaching the CNC machine. Prior to the part making it on to the machine, the fabrication team built a program that verified the toolpath developed by the engineer. The program provided data about the tool path and verified planned production run times for the team to utilise when planning a production schedule. In addition to building a production schedule, the program provided other information to the team allowing them to run parts through the night without supervision.

4. Applying coating to milled foam formwork.

5. Rebar placement on formwork actual.

6. Access Anvil's Concrete Form Post Coating. Photo: Timothy Schenck

7. The Fort Miller Company's Production Line Assembling Rebar on the Concrete Form. Photo: Timothy Schenck.

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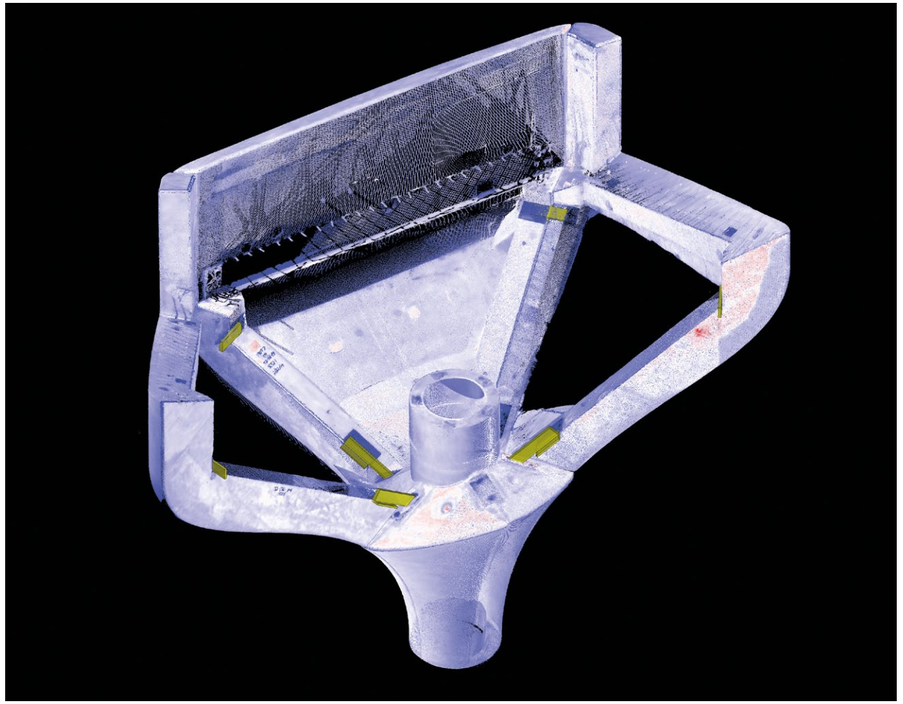

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New Surface Treatment
After the parts were milled and assembled, the surface aesthetic was applied. Traditional products such as architectural grade plywood or formliners are utilised to achieve the concrete finishes on various construction projects. Due to the nature of the shapes and long lead times, a new method had to be implemented on a large scale that allowed the team to turn forms over quickly, achieve a surface aesthetic required by the design team, and be used repeatedly for design economy. A high pressure polyurethane product was applied to the foam that met all of the design criteria.

Automation of Shop Drawings Generation
The development of the form was the first challenge to overcome for the fabrication team, and the challenges continued to pressure traditional construction methods. Parametric programming software allowed the team to address other issues associated with the construction of the concrete shapes. The custom programs and scripts generated by parametric programs automated the process of creating 2D cross sections and views from 3D models of each pot to be used in the development of precast concrete shop drawings. Tables were developed for each rebar used in the production process from the 3D scripts that were generated from the design team's scripts. The net effect of the scripts to produce shop drawings significantly reduced the amount of engineering time to develop 2D drawings manually for production. Data from the scripts had to be translated in an understandable way for the manufacturing purposes. The use of custom programs converted the massive amounts of data into readable .txt and .csv files for manufacturing.

Digital Information Transfer
Manual methods of digital information transfer were not feasible. Equipment acquired had to speak the same language relative to the 3D models. The fabricators relied heavily on the tolerances of the CNC equipment to mill the forms, automatically bend the rebar and laser-cut the various embed plates connecting petal to petal within a pot and unique steel geometry for stay-in-place formwork to hold the pots together. The sharing of information and connectivity between three different companies allowed the fabrication team to trust and verify each part was built to tolerance, would fit in its location and be set as needed, a unique communication experience.

8. Virtual assembly of 3D scanned precast elements.

9. Pier 55 during Construction. Photo: Timothy Schenck.

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Quality Control Using Digital Tools
The quality control procedures also relied on digital processes. Since the architectural shapes were undulating and not orthogonal, we digitally scanned the as-built part, downloaded the scan into CAD software and overlaid the scan on the 3D model to verify the results of the part compared to the 3D model. The process provided a sense of tolerance control and further verification of the process.

Construction

Assembly
All concrete and steel components fabricated at Fort Miller Company and Access Anvil were transported by lorry the one-hour drive to the assembly site at Port of Coeymans on the Hudson river edge. A total of eight assembly stations were set up by the marine contractor to meet the tight schedule. Completely assembled pots were placed into the hopper barge that could store a maximum of four pots. A total of four barges operated in turns for delivery of the assembled pots to the project site.

Site Erection
Pots arrived at the site in the barge and were dragged by tug boat to be erected by a floating crane on to piles. A well-planned erection procedure meant one pot was erected in a few hours.

Topping Slab
Finally, the cast-in-place topping slab was poured to tie all the pots and the flat pier together and create a monolithic, highly indeterminate moment-frame pier structure.

Conclusion

Little Island is a ground-breaking departure from conventional pier construction and utilises the latest advances in digital design and fabrication technology to successfully deliver a structure that is both functional and playfully sculptural at the same time. The major challenge of generating, engineering, communicating and fabricating the complex geometry was overcome with intensive use of parametric scripts, automation and digital data communication in the design phase, and the use of milled foam formwork and bending rebar using a detailed 3D model in the fabrication phase. The collaboration between all parties to deliver this project was critical in achieving a successful outcome for future generations to enjoy. When complete, Little Island will be a new icon of the west side, and a visible reminder of what can be achieved when the conventional scoping boundaries of contractors and designers are blurred and project participants come ready to collaborate.

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of Stuttgart, he focuses on robotically-fabricated lightweight composite structures.

Moritz Dörstelmann is managing partner of FibR GmbH (www.fibr.tech), a specialist company for computational design and robotic fabrication of bespoke fibre composite structures, which enables the exploration of a novel design and construction repertoire for expressive high-performance lightweight structures. His work on digital fabrication technology provides societally relevant solutions for resource efficient architectural construction. Dörstelmann is a registered architect. He developed the underlying digital design and robotic fabrication strategies of his work through seven years of research at the University of Stuttgart, Harvard University and as visiting professor for Emerging Technologies at the Technical University of Munich.

Ondrej Kyjanek is co-founder and CTO of FibR GmbH, a specialist company of bespoke filament structures, and a former research associate at the Institute for Computational Design and Construction (ICD) at the University of Stuttgart, where his research focused on advanced robotic automation and human-machine collaboration in the timber construction industry. He studied architecture and urban planning at the Czech Technical University in Prague (Bc.), followed by the Integrative Technologies and Architectural Design Research (ITECH) programme at the University of Stuttgart (MSc.).

Jan Knippers See page 312.

Achim Menges See page 312.

4: POLEMICAL PERFORMATIVE PRACTICE

Embedded Architecture: Ada, driven by humans, powered by AI

Jenny E. Sabin See Editor's Biographies page 310.

John Hilla is a Designer with Jenny Sabin Studio and a Research Associate in the Sabin Design Lab at Cornell AAP. As a graduate of the Master of Architecture programme at the University of Pennsylvania, Hilla was awarded the Henry Adams Medal and was a finalist for the Student of the Year honour from the American Institute of Architects. Additionally, he received the Outstanding Graduate Student Award and the Legacy Award for Academic Engagement at Syracuse University, where he earned a Master of Science degree in Engineering Management and a Bachelor of Science degree in Mechanical Engineering.

Dillon Pranger is a licensed architect and senior designer at Jenny Sabin Studio. His current research involves unconventional building materials and construction techniques. These explorations have been realised through his leadership roles on various Jenny Sabin Studio pavilion installations, including Ada for Microsoft Research, Lustre for House of Peroni 2018, and Lumen for the 2017 MoMA PS1 Young Architects Programme. In addition to practicing, Pranger has taught architecture at various institutions, including Harvard GSD, Cornell University, and Syracuse University. He holds a B.S. in Architecture from the University of Cincinnati (2012) and a M.Arch from Cornell University (2015).

Clayton Binkley trained as a sculptor and a structural engineer and practices as an artist, craftsman and multidisciplinary designer. He received his BA in sculpture from Yale University and his MEng in structural engineering from the University of Bath. Binkley has collaborated extensively with Jennifer Newsom and Tom Carruthers of Dream The Combine, most recently on the 2018 MoMA PS1 Young Architects Programme installation, Hide & Seek. Working with Arup for over a decade, he has used his skills as a structural engineer to help open up new possibilities for artists and designers, including: Janet Echelman, Jenny Sabin, John Grade, Lead Pencil Studio and Jill Anholt.

Jeremy Bilotti is a graduate fellow at MIT's Design and Computation Group, and a dual-degree candidate in MIT's Department of Computer Science. He is also leading research projects at MIT's Self-Assembly Lab. As a designer at Jenny Sabin Studio, Bilotti led the development of software and hardware systems in collaboration with Microsoft Research for Ada. During his B.Arch at Cornell University, he was Senior Research Associate at the Sabin Design Lab, and contributed to groundbreaking work at Cornell's Robotic Construction Laboratory. Bilotti's work has been published by Routledge, RobArch, in ACADIA's peer-reviewed proceedings, and was exhibited in the 2019 Shenzhen Biennale.

Diagrams of Entropic Forces: Design for new dissipative fabrication

Philip Beesley is a multidisciplinary artist and architect. His research is recognised for its pioneering contributions to the field of responsive interactive architecture. Beesley directs Living Architecture Systems Group, an international group of researchers and creators. He is a professor at the University of Waterloo, Ontario and the European Graduate School. He represented Canada at the 2010 and 2020 Venice Biennale of Architecture. Beesley's collaborations with artists, scientists and engineers include diverse projects from haute couture to complex electronic systems that can sense, react and learn. This experimental work explores subtle phenomena and constantly changing boundaries at the outer edges of current technology.

Discrete Timber Assembly

Gilles Retsin is an architect and designer, who founded Gilles Retsin Architecture. He studied architecture in Belgium, Chile and the UK, where he graduated from the Architectural Association. His design work and critical discourse has been internationally recognised through awards and exhibitions at institutions including the Museum of Art and Design, New York and the Centre Pompidou, Paris. Retsin edited an issue of *Architectural Design* on the Discrete and has co-edited *Robotic Building: Architecture in the Age of Automation*. He is Programme Director of the M.Arch Architectural Design at The Bartlett, UCL. He co-founded the UCL Design Computation Lab which does high profile research into new design and fabrication technologies.

Towards Discrete Automation

Mollie Claypool is an architecture theorist and Co-Director of Design Computation Lab (DCL) at The Bartlett School of Architecture, UCL in London where she is also Coordinator of History & Theory in MArch Architectural Design. She is Managing Editor of *Prospectives*, a new open-source publishing platform to be launched in 2020 for research on computation and automation in architecture and design. She is Director of Automated Architecture (AUAR), a design and technology startup based in the UK and is co-author of the recent book *Robotic Building: Architecture in the Age of Automation* (DETAIL Edition, 2019). Claypool was awarded a Fellowship in Automation by the South West Creative Technology Network, funded by Research England, in 2019.

House #05: Incremental construction in digital practice

Paul Loh is an architect and senior lecturer at the Melbourne School of Design, University of Melbourne. His research focuses on the cognitive engagement of making in design practice with digital fabrication and robotics. He studied architecture at the University of Melbourne, University of East London (UEL), the Architectural Association (Design Research Lab) and gained his doctorate at RMIT University with a dissertation on Digital Material Practice: the Agency of Making. Loh has previously taught at UEL and the AA. He is a founding partner of LLDS / Power to Make.

David Leggett is a registered architect in Australia and the UK. Leggett studied architecture at the University of East London and the University of Westminster. He is a founding partner of LLDS / Power to Make, a new breed of architectural practice that integrates digital fabrication in the design of architecture. He established the Architectural Research Laboratory in 2017, which explores novel fabrication techniques and processes applicable to the construction industry. Leggett teaches graduate design Studio 15 at the Melbourne School of Design.

Concrete Choreography: Prefabrication of 3D-printed columns

Ana Anton is a PhD Candidate at the chair for Digital Building Technologies, ETH Zurich and associated to the National Centre for Competence in Research – Digital Fabrication, where she leads the research in concrete 3D printing. She received her architectural degree, cum laude, at TU Delft, in 2014 and continued her research until 2016 as part of the Hyperbody Research Group. While her scientific research addresses complexity and emergence in architecture, her designs exploit materiality encoded for digital fabrication. Her current thesis, 'Tectonics of Concrete Printed Architecture', focuses on robotic concrete extrusion processes for large scale building components.

Patrick Bedarf is a PhD researcher at the chair for Digital Building Technologies, Institute of Technology in Architecture at the Department of Architecture of ETH Zurich and recipient of the ETH research grant. After studying architecture at HTWK Leipzig and TU Delft, he gained experience internationally in offices including Zaha Hadid (London), Delugan Meissl (Vienna) and Henn (Berlin, Beijing). Bedarf lectured and tutored workshops in architectural schools throughout Europe (ETH Zürich, TU Delft, TU Berlin, Politecnico di Milano). His work challenges conventional building culture through innovative design and fabrication processes and currently investigates large-scale 3D printing with porous materials.

Angela Yoo is currently a Research Assistant at the chair of Digital Building Technologies, Institute of Technology in Architecture of ETH Zurich. Upon graduating with a Master's of Architecture (Prof. Hons) from the University of Auckland in 2014, she gained professional experience at Monk Mackenzie Architects based in New Zealand. Moving to Switzerland in 2017, she graduated from the MAS in Architecture and Digital Fabrication at ETH Zurich and has since been involved in research and teaching activities. Her recent projects have focused on design for robotically 3D-printed concrete and 3D-printed formwork for functional building components in architecture.

Lex Reiter is a post-doc researcher at ETH Zurich in the group Physical Chemistry of Building Materials. He works on early age strength build-up and its control for concrete extrusion 3D printing and similar digital fabrication processes with concrete. His research interest is in the physical and chemical processes that enable concrete to be shaped without formwork at high vertical rate, as well as the associated processing challenges.

Timothy Wangler is a senior researcher in the Physical Chemistry of Building Materials group, Institute for Building Materials, at ETH Zurich. He received his PhD in Chemical Engineering from Princeton University with an emphasis on organic-inorganic interactions in construction materials, with his recent research focusing on materials and processing applications in digital fabrication with concrete.

Robert J. Flatt became Professor of Physical Chemistry of Building Materials at Institute of Building Materials, ETH Zurich in September 2010. Before that he was Principal Scientist Head of Inorganic Materials in Corporate Research at Sika Technology AG, the R&D branch of Sika AG. Prior to joining Sika, he spent two and a half years as postdoctoral researcher at Princeton University in the Department of Civil and Environmental Engineering within the materials group of Prof. G.W. Scherer. He studied at the Swiss Federal Institute of Technology in Lausanne, where he obtained his masters in Chemical Engineering and a PhD from the Materials Science Department.

Architect **Benjamin Dillenburger** is Assistant Professor for Digital Building Technologies at the Institute of Technology in Architecture at the Department of Architecture, ETH Zurich. His research focuses on the development of building technologies based on the close interplay of computational design methods, digital fabrication and new materials. In this context, he searches for ways to exploit the potential of additive manufacturing for building construction. He previously was appointed as Assistant Professor at the John H. Daniels Faculty of Architecture, Landscape and Design at the University of Toronto. He obtained his PhD and worked as a senior lecturer in the CAAD group, at ETH Zurich.

Little Island: Sculptural structure through digital design and fabrication

Yong-Wook Jo is an Associate in the New York office of Arup, and leads the firm's multi-disciplinary team as the project manager for Little Island project. Trained in both architecture and engineering, he has collaborated with the world's top architects and contractors with shared understanding of iconic projects featuring complex and irregular geometries and unique technical challenges. Jo's portfolio includes retail, tall buildings, education, airport as well as marine and infrastructure projects. He received a bachelor's degree in architecture from Yonsei University, Korea, and a master's degree in structural engineering from the University of California, Berkeley.

Rosario Gallo leads the team at Access Anvil Corp. (AAC), a Fort Miller Group company, located in Greenwich, New York. AAC is an ISO 9001 certified company that specialises in the design and fabrication of custom concrete formwork and five-axis CNC milling techniques. In addition to its fabrication and milling capabilities, AAC acquired the Scott System product line located in Denver, Colorado to add fabrication of urethane formliners, plastic formliners and brick inlay products to its concrete forming line. Our team provides innovative solutions to complex construction problems.

David Farnsworth is a Principal with Arup's New York office. Farnworth has a depth of experience in the design and management of projects both locally and globally. His experience includes tall buildings, piers, bridges, rail, airport and civil projects. Having successfully delivered projects through all types of procurement methods – including design bid build, design build, construction management, and direct work with contractors – Farnworth enjoys collaborating with all project participants to deliver innovation and risk mitigation to each new project. He currently leads Arup's Americas Property Market Business and has designed some of the world's most iconic tall building structures.

Jacob Wiest is a Senior Engineer at Arup who has spent time in their New York and Sydney building structures teams, working on projects varying in typology and market. His proficiency in parametric and digital tools supports an interest in projects with complex geometry and data management requirements. Wiest attended the Pennsylvania State University where he received a Master's and Bachelor's of Architectural Engineering.

Hudson Yards Vessel, New York

Edoardo Tibuzzi See page 314.

Jeg Dudley See page 314.

Pablo Zamorano is Head of Geometry and Computational Design at Heatherwick Studio. He works across all projects providing expertise and guidance on new technologies and techniques, and the execution of challenging geometries. Since joining the Studio in 2015, Zamorano has also worked as Deputy Project Leader on Coal Drops Yard, an award-winning retail quarter and public space in King's Cross, London, where he oversaw all design packages into construction. Prior to working at the studio, Zamorano worked at SOM London, as well as New York City and Santiago. He has lectured widely, and his personal work has been published and awarded internationally.

Peter Romvari joined Heatherwick Studio in 2011. Since then, Romvari has made a significant contribution to a number of the studio's projects including Vessel – the landmark centrepiece for Hudson Yards in Manhattan's West Side. He played a crucial role in driving the overall geometry of the highly complex structure, from concept phase through to informing fabrication and construction. Most recently, Romvari has worked on a kinetic glasshouse, developing the geometry for a project that is as much a machine as a building. His background is architectural practice and the visual arts, having worked as a 3D artist at Atelier Peter Kis.

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